OFFICIAL TECHNICAL SPECIFICATION DATASHEET
MANUFACTURER: TRINA SOLAR
MODEL / SERIES: Vertex S+ TSM-440NEG9R.28
DOCUMENT TITLE: Trina Solar Vertex S+ Dual-Glass TSM-NEG9R.28 Datasheet
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1. PRODUCT IDENTIFICATION & ARCHITECTURE

Cell Technology:	N-type TOPCon
Module Construction:	Bifacial (Glass-Glass)
Number of Cells / Format:	144 (half-cut) N-type i-TOPCon Cells
Glass / Front Encapsulant:	1.6 + 1.6 mm High Transmission Dual Glass
Frame Material:	Anodized Aluminium Alloy (Black Frame)
Junction Box / Protection:	IP68 with 3 bypass diodes

2. ELECTRICAL CHARACTERISTICS AT STC (1000 W/m2, 25C, AM 1.5)

Nominal Maximum Power (Pmax / Pnom):	440.0 Wp
Module Aperture Efficiency:	22.00 %
Open Circuit Voltage (Voc):	52.20 V
Short Circuit Current (Isc):	10.67 A
Maximum Power Voltage (Vmp):	44.00 V
Maximum Power Current (Imp):	10.01 A
Maximum System Voltage:	1500.0 V DC
Maximum Series Fuse Rating:	25.0 A

3. TEMPERATURE CHARACTERISTICS & OPERATING LIMITS

Temp. Coefficient of Pmax (gamma):	-0.290 %/C
Temp. Coefficient of Voc (beta):	-0.240 %/C
Temp. Coefficient of Isc (alpha):	+0.040 %/C
Nominal Cell Temperature (NOCT/NMOT):	43 +/- 2 C (NOCT)
Operating Temperature Range:	-40 C to +85 C

4. MECHANICAL DIMENSIONS & RATINGS

Length:	1762.0 mm
Width:	1134.0 mm
Thickness (Height):	30.0 mm
Gross Aperture Area:	1.9981 m2
Module Weight:	21.0 kg
Mechanical Design Load (Front):	5400 Pa (Snow load)
Mechanical Design Load (Rear):	2400 Pa (Wind load)

5. WARRANTY & LIFETIME PERFORMANCE

Product Workmanship Warranty:	25.0 Years
Linear Power Output Warranty:	30.0 Years
Annual Linear Degradation:	0.40 % / year
End of Warranty Retained Output:	87.4 %

6. PROVENANCE & ARCHIVAL METADATA

Authoritative URL:	https://static.trinasolar.com/sites/default/files/Datasheet_VertexS_Plus
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SolarSelect MLOps Hardware Database Reference Standard IEC 61215 / IEC 61730